

# Beckett Frey

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## EDUCATION

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**UW Milwaukee | Milwaukee, WI, USA**

*B.S. Computer Science | January 2023 - May 2025*

- Dean's Honor List, Fall 2024 & Spring 2025

## TECHNICAL SKILLS

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**Programming Languages:** Python, TypeScript, Bash, SQL

**Tools and Frameworks:** Django, React, PyQt6, LangGraph, Node.js

**Cloud and Infra:** GitHub Actions, Docker, Modal, AWS

**Machine Learning:** Speech (Whisper, Pyannote, Forced Alignment), Vision (YOLO, Synthetic Data)

## WORK EXPERIENCE

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### Software Consultant & Engineer (Contract)

*Independent, Madison, WI, USA | June 2026 - Present*

- Performed application security review of a production SaaS product and delivered prioritized written recommendations to the client.
- Coordinated an independent third-party security assessment on the client's behalf, from firm evaluation through scoping and engagement.
- Built out separate development and staging infrastructure to move code changes out of production workflows.
- Advise a prospective startup founder on technical feasibility, architecture, and MVP scoping.

### Research Intern

*Waisman Center, WI, USA | September 2025 - September 2026*

- Built and shipped a cross-platform desktop application for phonetic research, reducing onboarding friction and improving metadata tracking for reproducible workflows.
- Developed tooling supporting machine learning workflows for automated pronunciation assessment in children with speech sound disorders.
- Built CI/CD pipelines with GitHub Actions and Docker across multiple repositories, improving reliability and reducing maintenance overhead for researchers.
- Collaborated across labs to support the ongoing development of legacy Django infrastructure, focusing on modernization and knowledge continuity.

### Software Engineer (Volunteer)

*Environmental Data & Governance Initiative, Remote, USA | January 2026 - Present*

- Resolve bugs and deliver features for the web-monitoring project in response to analyst feedback.
- Improve CI with ruff and git hooks to gate low-quality open-source contributions before review.

### Junior ML Engineer

*Hyperchess.ai, Remote, USA | December 2024 - September 2025*

- Built a synthetic data pipeline and a 25-model YOLO mixture-of-experts for piece detection, reaching 99% mAP@0.5.
- Deployed inference workloads with Modal serverless GPU, drastically improving the speed of iteration.

## PROJECTS

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- **Phonetic Research Application**, Cross-platform desktop app that unifies AI/ML tools (MFA, Wav2TextGrid, Faster-Whisper, PyPLLRComputer) behind one interface, letting Speech Language Pathology researchers build and run phonetic analysis pipelines in one place. To be presented at Interspeech 2026. [voxbkit-web.vercel.app](https://voxbkit-web.vercel.app)
- **AI Document Generation**, Python CLI and library that fills markdown templates with LLM-generated content under strict length and structure constraints, using batched context-aware generation and a LangGraph morphing loop to enforce character bounds. [github.com/BeckettFrey/doc-weaver-agent](https://github.com/BeckettFrey/doc-weaver-agent)
- **Data Security Tooling**, Published PyPI tool that scans git repositories for PII and other identifier patterns. Used to safely open-source several lab repos and caught one real PII exposure in legacy code. [pypi.org/project/shred-guard](https://pypi.org/project/shred-guard)

## EXTRACURRICULAR ACTIVITIES

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- **Open Source Contributor**, Github - Ongoing
- **Inpatient Unit Support**, UW Health - Ongoing
- **Triathlete**, IRONMAN Wisconsin - September 2023